AI Teaching Assistants in Asynchronous Discussion Forums: A Mixed-Methods Study of Students' Community of Inquiry Presence

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Abstract

Grounded in the Community of Inquiry (CoI) framework, this study examined how an artificial intelligence teaching assistant (AI TA) shaped students' perceived and behaviorally enacted CoI presence in asynchronous discussion forums (ADFs). TA support was examined in a 10-week blended course through a three-stage experimental cohort design, including no-TA, human-TA-supported, and AI-TA-supported discussion stages, together with a historical control cohort receiving continuous human-TA support for behavioral comparison. An explanatory mixed-methods design was adopted, combining CoI questionnaires, coded student-authored forum posts, forum-based reflections, and semi-structured interviews. The results revealed a perception–behavior gap: students reported higher cognitive presence (CP), social presence (SP), and teaching presence (TP) during the AI-TA-supported stage, but these perceived improvements were not fully mirrored in observable forum behaviors. Behaviorally enacted CP increased most clearly during the human-TA-supported stage in the experimental cohort and showed no further significant increase during the AI-TA-supported stage, while behaviorally enacted SP and learner-enacted TP remained sparse in student-authored posts. Qualitative findings suggested that, compared with human TA support, the AI TA was perceived as advantageous in providing structured cognitive support, a low-pressure interaction atmosphere, and timely and accessible guidance, but limited in inquiry flexibility, emotional authenticity, and supervisory authority. These findings highlight the role boundaries of AI TAs in ADFs: they may strengthen students' perceived support, but their contribution to deeper inquiry and peer interaction depends on human orchestration and complementary facilitation. AI TAs should therefore be integrated into human–AI collaborative facilitation models that preserve the complementary roles of human TAs and instructors.

Keywords: artificial intelligence teaching assistant; asynchronous discussion forums; Community of Inquiry; pedagogical agents; mixed-methods

Introduction

Asynchronous Discussion Forums (ADFs) serve as a cornerstone of online and blended learning, with wide application in Massive Open Online Courses (MOOCs) and Small Private Online Courses (SPOCs). By fostering learning communities and providing interactive educational experiences, ADFs enable students to develop disciplinary expertise and critical thinking skills through engagement with peers and instructors (Robbins & Fairbank, 2023). The asynchronous nature of these forums provides learners with ample time for reflection (Hull & Saxon, 2009), allowing them to formulate inquiries, craft considered responses in online discussion spaces (Miyashita & Wark, 2024), and engage in collaborative knowledge construction and higher-order critical thinking (Galikyan & Admiraal, 2019). However, due to the finite time and energy of instructors, providing timely and meaningful feedback for the vast volume of posts remains a significant challenge. Coupled with the inherent latency in peer interactions, students often experience insufficient discussion feedback, which adversely affects their learning experience and overall outcomes. While prior research has attempted to mitigate this through strategies such as setting up teaching assistants (Yang, 2008), optimizing instructor-student ratios (Parks-Stamm et al., 2017), and implementing reply-priority systems (Irish et al., 2023), these approaches rely heavily on limited human resources and cannot fully resolve the need for timely and sustained facilitation in ADFs.

Pedagogical Agents (PAs) offer a potential approach to addressing these challenges. PAs can be designed with course-specific knowledge, theory-informed interaction logic, and API-based integration to provide automated and timely feedback in online learning environments. A few studies have deployed agents as teaching assistants (TAs) on online learning platforms. The most prominent example is Khanmigo, an AI-powered tutoring chatbot developed by Khan Academy, which offers Socratic guidance to students while assisting teachers with instructional tasks (Khan Academy, *n*.d.). Another notable example is Jill Watson, an AI virtual teaching assistant developed by Georgia Tech's Design Intelligence Lab (Design Intelligence Lab, *n*.d.). Jill Watson can answer inquiries regarding course syllabi and engage students in meaningful conversations about course materials with enhanced accuracy and reduced hallucinations (Taneja et al., 2024).

Although ADFs serve as a cornerstone of online and blended learning and support reflection, peer interaction, and collaborative knowledge construction, the application of PAs in ADFs remains underexplored. In particular, existing PAs have primarily been used for individual tutoring, question answering, or platform-based learning support, whereas their role in facilitating ongoing peer discussion in ADFs has received limited attention. This study addresses this gap by designing a PA that functions as an artificial intelligence teaching assistant (AI TA) within ADFs and participates in the ongoing discussion process. Given that cognitive presence (CP), social presence (SP), and teaching presence (TP) are considered key indicators of meaningful online learning and substantially influence students' learning experiences and discussion outcomes in online and blended learning environments (Martin et al., 2022), the Community of Inquiry (CoI) framework was adopted to guide the design and evaluation of the AI TA. The study then employed an explanatory mixed-methods design to examine students' self-reported CoI presence, behaviorally enacted CoI presence in forum interactions, and perceptions of the AI TA under different TA-supported discussion conditions. The specific research gaps and research questions are presented after the review of relevant literature.

Literature Review

CoI Presence and Facilitation in Online Discussion Forums

The CoI framework explains how CP, SP, and TP interact to support meaningful educational experiences and learning processes in online environments (Garrison et al., 1999) (see Figure 1). In online education, the CoI framework serves as a theoretical foundation for constructing learning environments and evaluating online learning experiences (Akyol & Garrison, 2008). Higher perceived levels of these three presences have been associated with better course grades, student experience, perceived learning, and satisfaction (Rockinson-Szapkiw et al., 2016; Stenbom, 2018).

CP refers to the degree to which learners "collaboratively construct meaning through sustained discourse within a CoI" (Garrison et al., 2001). It is commonly analyzed through the

Practical Inquiry Model, including triggering events, exploration, integration, and resolution (Garrison et al., 1999). A persistent challenge in asynchronous discussions is that cognitive activity often stalls at the exploration phase, rarely reaching higher-order integration or resolution (Garrison & Arbaugh, 2007; Naeem & Khan, 2019). This suggests that merely providing an online space for discussion is insufficient; students often need strategic prompts, questioning, synthesis, and guidance to move from information exchange toward deeper inquiry (Darabi et al., 2011; Koh et al., 2010).

SP is defined as the "capacity of participants in a CoI to project themselves socially and emotionally as 'real' people" (Garrison et al., 2010). It serves as the foundation for a collaborative learning community by supporting affective expression, open communication, and group cohesion. In ADFs, SP is especially important because students interact without immediate face-to-face cues and must rely on textual communication to establish interpersonal connection, trust, and group belonging. However, sustaining SP remains challenging when peer responses are delayed, participation is uneven, or students perceive the forum as a task-completion space rather than a learning community.

TP involves the "design, facilitation, and direction of cognitive and social processes" to achieve meaningful learning outcomes (Anderson et al., 2001). It is enacted through instructional design, discourse facilitation, direct instruction, and activities such as clarifying expectations, encouraging participation, maintaining discourse, asking probing questions, and providing feedback (Garrison, 2007). Although TP is often initiated by instructors, it may also be distributed among instructors, teaching assistants, and students during online discussions.

However, because instructors and human teaching assistants have limited time and attention, providing timely and sustained facilitation for every student post remains difficult, especially in forums with large volumes of discussion.

In summary, the CoI framework underscores that ADFs can support reflection, peer interaction, and collaborative knowledge construction, but these outcomes depend heavily on effective facilitation. These challenges provide the pedagogical basis for examining technology-supported facilitation in CoI-oriented ADFs.

Pedagogical Agents in Online Learning

Pedagogical Agents (PAs) are defined as life-like characters or software interfaces designed to facilitate learning in computer-mediated environments (Kim & Baylor, 2006). Traditionally, PAs served as demonstrators or tutors within structured learning systems, focusing on delivering content or guiding students through predefined learning paths. With the integration of natural language processing and large language models (LLMs), PAs have gradually shifted from scripted and presentation-oriented agents to more conversational, adaptive, and generative learning partners (Lee et al., 2024). A recent systematic review shows that AI chatbots can provide personalized learning support, immediate feedback, homework assistance, skill development, and time-saving pedagogical assistance, while also raising concerns about accuracy, reliability, transparency, ethical use, and human oversight (Labadze et al., 2023).

In online learning environments, AI agents and PAs have been used to support several instructional and interactional functions. First, they can serve as tutors or content presenters by explaining concepts, modeling procedures, and guiding learners' attention to key information;

these roles are consistent with broader reviews of pedagogical agent design and online AI learning support (Labadze et al., 2023; Zhang et al., 2024). Second, AI agents can provide feedback, scaffolding, and formative guidance during learning activities. Studies of educational chatbots and online discussion agents show their potential to provide immediate, personalized, and contextually coherent support that helps learners interpret information, monitor progress, and participate in online learning activities (Labadze et al., 2023; Li et al., 2022).

Third, AI agents may support affective and social dimensions of online learning. Affective and chatbot-based agents can provide emotional support, immediate responses, and social cues, while empathic chatbot research suggests that machine expressions of sympathy and empathy can shape users' perceptions of social and emotional support (Labadze et al., 2023; Liu & Sundar, 2018). Finally, AI agents can be embedded in broader platform-based learning environments to support multiple roles, including teaching, peer interaction, co-regulation, and learning guidance, but such support still requires careful human-centered design and oversight (Li et al., 2022; Ouyang & Jiao, 2021; Zawacki-Richter et al., 2019).

Taken together, existing studies indicate that AI agents can support online learning through tutoring, feedback, scaffolding, affective support, learning guidance, and platform-based interaction. However, much of this work has focused on individual learner-agent interaction, video-based instruction, formative assessment, or general platform support. Comparatively less is known about how an AI agent can be embedded in asynchronous discussion forums, where learning depends not only on individual support but also on sustained peer interaction, discourse facilitation, and collaborative inquiry.

Toward CoI-Oriented AI TA Facilitation in ADFs

The functions of AI agents reviewed above suggest that they may be used to support several processes that are central to CoI-oriented facilitation in ADFs. From the perspective of CP, recent research demonstrates that GenAI can support students' cognitive engagement when used as a structured scaffold. For instance, Yu et al. (2025) integrated GenAI into six-hat thinking scaffolded discussions, where students consulted GenAI for information retrieval, idea generation, and problem-solving during instructional design tasks. Their findings suggested that GenAI-supported groups generated more discussion posts and showed stronger connections among higher-level CP indicators, particularly among high-creativity pre-service teachers. However, in that study, GenAI functioned mainly as a student-initiated cognitive tool rather than an embedded facilitator that directly participated in forum interactions.

From the perspective of SP, AI assistants may provide human-like social cues at scale. Research on AI interfaces and customer-service chatbots suggests that anthropomorphic design cues and immediate responses can enhance users' perceived social presence and improve their attitudes toward the interface or brand (Rao Hill & Troshani, 2024; Williams, 2025). However, these chatbots mainly function as one-to-one service or interface agents, aiming to create interpersonal warmth, responsiveness, and user engagement, rather than to facilitate peer interaction or group discourse in collaborative learning communities.

From the perspective of TP, AI agents may also enact certain facilitation and instructional support functions. Wang et al. (2024) found that learners perceived cognitive, social, and teaching presences when interacting with an AI coach, and that teaching presence negatively predicted learning outcomes. They suggested that this result may reflect limitations in the design and implementation of AI support rather than the inherent inability of AI agents to facilitate learning. This finding is especially relevant for ADFs because AI-based teaching support may be perceived by students as useful or present, but such perceptions do not necessarily indicate that AI support effectively promotes peer interaction, sustained inquiry, or improved learning outcomes.

Together, prior studies suggest that AI agents may provide social cues, cognitive scaffolding, and instructional guidance for online learning, while CoI-oriented ADF research has mainly fostered online discussion through human facilitation strategies, such as instructor modeling, discussion prompts, feedback, participation requirements, and teaching assistant support (Anderson et al., 2001; Darabi et al., 2011; Koh et al., 2010; Yang, 2008). However, embedding an AI TA in ADFs differs from using AI chatbots for individual tutoring, customer-service interaction, or student-initiated cognitive support, because ADFs are collaborative learning spaces in which knowledge construction depends on sustained peer interaction, shared inquiry, and the progressive development of discussion threads. Although emerging studies on chatbots, GenAI tools, and pedagogical agents suggest the potential of AI-supported facilitation to provide social cues, cognitive scaffolding, and instructional guidance (Rao Hill & Troshani, 2024; Wang et al., 2024; Williams, 2025; Yu et al., 2025), few studies have examined whether such agent-based support can be embedded into ADFs to facilitate all three dimensions of CoI presence during ongoing discussion. This gap highlights the need to evaluate AI TA support not only through students' self-reported CoI presence, but also through their behaviorally enacted

CoI presence in actual forum interactions and their qualitative experiences of AI-supported facilitation.

Research Gap and Present Study

Taken together, prior CoI research has established CP, SP, and TP as interrelated constructs for understanding meaningful online and blended learning experiences, including text-based discussion environments (Redstone et al., 2018), while research on PAs, especially pedagogical AI conversational agents, has shown their potential to provide cognitive guidance, affective support, feedback, and learning support in online learning environments (Yusuf et al., 2025). However, these two research streams remain insufficiently connected in ADFs. Limited evidence is available on how PAs, when embedded as AI TAs, participate in ADF interactions and shape students' CoI presence. To address this gap, the present study designed an AI TA for ADFs based on CoI-oriented facilitation principles and examined its role in students' CoI presence across different TA-supported discussion conditions. Rather than evaluating AI TA support only through students' subjective perceptions, this study also considered how CoI presence was behaviorally enacted in actual forum interactions, thereby providing a more comprehensive account of AI-supported facilitation in ADFs. The study was guided by the following research questions:

RQ1: How does students' self-reported CoI presence vary across no-TA, human-TA-supported, and AI-TA-supported discussion stages?

RQ2: How does students' behaviorally enacted CoI presence vary across discussion stages and between the experimental cohort and the historical control cohort?

RQ3: What are students' overall perceptions of the AI TA, and how do they perceive its advantages and limitations compared with human TA support in ADFs?

Design and Implementation of the AI TA

This section describes how the AI TA was designed and implemented as an embedded facilitator in ADFs, including its CoI-oriented interaction strategies, response-generation pipeline, and examples of forum interaction.

CoI-Oriented Interaction Design

Prior research suggests that instructional strategies effective for human instructors may also remain effective when implemented through PAs (Zhang et al., 2024). Therefore, the AI TA's interaction design was adapted from established facilitation strategies (Table 1). The AI TA was designed as a form of TA support comparable to human TA support in the study design.

These strategies were integrated into a response-composition logic rather than implemented as isolated response functions. For initial responses to student posts, the AI TA followed a four-step pattern to provide prompt feedback, acknowledge students' contributions, offer cognitive scaffolding, and invite further peer discussion. This pattern guided the composition of AI replies rather than prescribing how the discussion itself should unfold.

Specifically, the AI TA first established conversational immediacy through prompt feedback, peer identity, and friendly salutation; second, affirmed and synthesized students' contributions through content summarization, emotional support, and personalized guidance; third, provided cognitive scaffolding through illustrative examples and critical challenges; and finally, invited the original poster and other peers to respond to, extend, or build on one another's ideas.

The four-step pattern was mainly applied to the AI TA's initial responses to student posts. In subsequent multi-turn conversations, the AI TA generated more concise and context-sensitive replies, reducing scaffolding while retaining necessary emotional support. This design was intended to avoid overly lengthy or mechanical responses that might reduce students' agency, interfere with peer-to-peer interaction, or weaken continued engagement.

System Implementation and Response Generation Process

The AI TA was developed on Coze (https://www.coze.com/), a generative AI agent development platform, and was connected to the ADFs through API-based integration. This integration allowed new student posts to be captured as input for backend processing and enabled generated replies to be returned through the AI TA forum account. As shown in Figure 2, the system consisted of two connected components: the forum-side interaction workflow and the AI TA response-generation backend.

On the forum side, the instructor initiated discussions by posting topics and guidance in the ADFs, while students participated by publishing original posts and replying to peers. When a new student post was submitted, it became visible to the AI TA and triggered the backend response-generation process. Instructor posts were treated as instructional prompts or guidance and did not trigger AI TA replies. After a response was generated, it was posted back to the forum through the AI TA account as an automated reply. In this way, the AI TA functioned as a visible forum-based teaching assistant, while the discussion space remained organized around instructor-initiated topics, student posts, and peer discussion.

On the backend side, response generation was organized through several functional modules. The perception module detected new posts, identified the author role, and extracted the post content, discussion topic, and immediate context. The memory module maintained short-term conversational context, while the database stored discussion logs, including student posts and AI-generated replies, to support log retrieval and context updating rather than individualized long-term learner profiling. The reasoning module, powered by DeepSeek-R1, integrated the post input, recent context, retrieved discussion records, course knowledge base, AI TA profile, and behavioral constraints to generate a response plan. The knowledge base provided course materials and instructional resources, while the profiling module specified the AI TA's identity setting, response workflow, and behavioral constraints. Auxiliary tools, including internet search, content safety checking, and link reading, were available when needed. Finally, the action module composed the final reply and returned it to the forum through the AI TA account.

Interaction Example

An example of AI TA interaction in the ADFs is presented in Figure 3.

Methodology

This study employed an explanatory mixed-methods design (Fetters et al., 2013) with a quasi-experimental structure and a historical control cohort (Walser, 2014). Quantitative data were used to examine students' perceived and behaviorally enacted CoI presence across different TA-support conditions, while qualitative data were used to further explain students' experiences with the AI TA and contextualize the quantitative findings.

Study Context and Participants

Data were collected from a blended graduate-level course, Educational Technology, at a university in southern China. The course combined face-to-face classroom teaching with after-class learning tasks, and ADF activities were integrated into the post-class learning phase.

The experimental cohort (Year N) consisted of 59 first-year master's students in educational technology, including 16 males and 43 females, who provided informed consent for their learning data to be used for research purposes. The historical control cohort (Year N−1) comprised 50 first-year students enrolled in the same course during the previous academic year, including 13 males and 37 females. Both cohorts were drawn from the same program at the same university, shared comparable academic backgrounds and admission criteria, followed the same course design, and were taught by the same instructor. The historical cohort therefore served as a behavioral comparison group under continuous human TA-supported conditions.

Research Design and Data Collection

As shown in Figure 4, the study followed a 10-week design combining a three-stage within-subject design in the experimental cohort with a historical control comparison. In the experimental cohort, students first participated in forum discussions without TA support during Weeks 1–2, followed by human-TA-supported discussions during Weeks 3–6 and AI-TA-supported discussions during Weeks 7–10. Self-reported CoI surveys were administered at the end of each stage, and forum discussion posts were continuously collected and coded to examine students' behaviorally enacted CoI presence. After the AI-TA-supported stage, 32 forum-based reflection posts and semi-structured interviews with six participants were collected to further explore students' interaction experiences with the AI TA.

The historical control cohort followed the same 10-week course structure and was supported exclusively by human TAs throughout the semester. Because the historical data were retrospective, only archived forum discussion posts were available, whereas self-reported survey data had not been collected in the previous academic year. These archived posts were used as behavioral comparison data for examining students' behaviorally enacted CoI presence across TA-support conditions.

Measurements

This study measured students' perceived CoI presence, behaviorally enacted CoI presence, and qualitative perceptions of the AI TA.

Self-Reported CoI Presence

Students' self-reported CoI presence was measured using the Community of Inquiry questionnaire developed by Arbaugh et al. (2008), which was adapted into Chinese for the present study. The instrument consists of 34 items rated on a 5-point Likert scale ranging from 1 = strongly disagree to 5 = strongly agree, including 12 items for CP, 9 for SP, and 13 items for TP. Cronbach's α coefficients were calculated separately for each dimension at each discussion stage. As shown in Table 2, the coefficients ranged from 0.82 to 0.92, indicating good internal consistency.

Behaviorally Enacted CoI Presence

Students’ behaviorally enacted CoI presence was examined through forum discussion posts collected across the discussion stages. The posts were analyzed using the CoI coding framework proposed by Garrison et al. (1999), which conceptualizes CP, SP, and TP through multiple discourse indicators. The operational definitions are shown in Table 3.

The behavioral CoI coding focused only on student-authored forum posts. Posts written by the instructor, human TAs, researchers, or the AI TA were treated as part of the instructional support conditions and were not included as coding units. This decision kept the behavioral analysis centered on students’ enacted CoI presence and avoided conflating students’ behavioral presence with the varying amount and form of facilitation provided under different TA-support conditions. To ensure comparability across stages and cohorts, the behavioral coding was based on selected comparable forum tasks rather than all forum activities. Topic-based opinion discussion forums and topic-based learning task forums were included because they required students to articulate viewpoints, discuss course-related issues, or respond to instructional prompts. Forums serving mainly administrative, introductory, or retrospective purposes, such as self-introduction forums and reflective summary forums, were excluded because they differed from regular discussion tasks in pedagogical function.

To establish the reliability of the behavioral coding, two trained coders independently coded the selected student-authored forum posts according to the behavioral CoI coding scheme. Inter-coder agreement exceeded 80% across the coded categories, and discrepancies were resolved through discussion until consensus was reached before the final coding results were used for analysis.

For CP, student posts were coded according to the four phases of the Practical Inquiry Model: triggering event, exploration, integration, and resolution (Garrison et al., 2001). In addition to phase frequency, a weighted behavioral CP score was calculated to distinguish lower-level inquiry from higher-order meaning construction, as online discussions often remain at the exploration phase and less frequently progress to integration or resolution (Garrison & Arbaugh, 2007; Garrison & Cleveland-Innes, 2005; Naeem & Khan, 2019). This scoring strategy did not assume that the four CP phases form an equal-interval linear scale; rather, it retained the frequency-based logic of traditional transcript analysis while introducing a conservative distinction between lower-level and higher-level inquiry. Triggering event and exploration were assigned a weight of 1, whereas integration and resolution were assigned a weight of 2:

$$CP = 1 \times n\text{Triggering Event} + 1 \times n\text{Exploration} + 2 \times n\text{Integration} + 2 \times n\text{Resolution}$$

For SP, because the selected forums were mainly topic-based opinion discussions and learning tasks, a strict coding standard was adopted to avoid overestimating SP from task-induced personal statements. Expressions of personal experience, preference, emotion, or attitude were coded as SP only when they explicitly functioned as interpersonal interaction, such as acknowledging another participant, responding to a peer's idea, expressing appreciation, showing affective alignment with others, or building a sense of group connection. In contrast, first-person statements that merely described one's own learning experience, viewpoint, preference, or task-related reflection were not coded as SP unless they were clearly directed toward another participant or the discussion community. This distinction was intended to ensure that behavioral SP reflected observable interpersonal and social interaction rather than the mere presence of personal pronouns or affective wording.

For TP, student posts were coded only when they reflected student-enacted teaching-presence-related behaviors, such as organizing discussion, facilitating peer exchange, focusing issues, or guiding collective inquiry. TP provided by instructors, human TAs, or the AI TA was excluded.

Qualitative Data on Students' Perceptions of the AI TA

To address RQ3, qualitative data were collected after the AI-TA-supported stage through forum-based reflections and semi-structured interviews. In the reflection activity, the instructor initiated a discussion thread with the prompt: "How do you perceive the performance of the AI TA? Has she helped you? What needs to be improved?" This activity yielded 32 reflection posts from the experimental cohort.

Semi-structured interviews were then conducted with six participants, including four males and two females, to obtain more in-depth insights into students' interaction experiences with the AI TA. The interview protocol was guided by the CoI framework and focused on students' perceptions of AI TA support for guidance and facilitation, social connection and participation, and thinking and understanding. It also included explicit comparative prompts asking participants to compare AI TA support with human TA support in terms of timeliness, accessibility, interaction pressure, emotional support, supervisory authority, and inquiry flexibility. Therefore, the qualitative data were used to understand both students' overall perceptions of the AI TA and their perceived advantages and limitations of AI TA support relative to human TA support across the three CoI dimensions. All interviews were audio-recorded and transcribed verbatim. The reflection posts and interview transcripts were then used together for qualitative analysis.

Data Analysis

Self-Reported CoI Presence

For the self-reported CoI questionnaire data, only students who completed all three stage surveys were included in the repeated-measures analysis ($n = 40$). One-way repeated-measures ANOVAs were conducted separately for CP, SP, and TP to examine differences across the no-TA, human-TA-supported, and AI-TA-supported stages. Shapiro-Wilk tests showed that CP and SP scores in the no-TA stage deviated from normality, whereas TP scores met the normality assumption across all stages. Mauchly's test indicated that sphericity was met for CP, $W = .99$, $\chi^2(2) = .37$, $p = .833$, violated for SP, $W = .53$, $\chi^2(2) = 24.12$, $p < .001$, and met for TP, $W = .98$, $\chi^2(2) = .73$, $p = .695$. Greenhouse-Geisser correction was therefore applied to SP. Bonferroni-adjusted pairwise comparisons, partial η^2, and Friedman sensitivity tests were reported where appropriate.

Behaviorally Enacted CoI Presence

Based on the forum selection and coding criteria described in Section 4.3.2, six comparable forum tasks were retained for the experimental cohort, with two from each stage, and three comparable tasks for the historical control cohort, with one from each stage. After excluding instructor prompts, human-TA/researcher feedback, AI-TA replies, and other non-student posts, the behavioral corpus included 300 student-authored posts from the experimental cohort and 154 from the historical control cohort. The full roster-based samples were retained, including 59 experimental and 50 historical-control students. If a student did not contribute analyzable posts in a selected forum or stage, the corresponding behavioral score was coded as zero.

After coding and aggregation, CP showed sufficient variation across students and stages, whereas SP and TP showed highly sparse distributions. Therefore, inferential analyses were conducted only for behavioral CP, while SP and TP were reported using descriptive statistics. Accordingly, assumption checks were conducted only for CP. Shapiro-Wilk tests indicated that CP scores deviated from normality across stages and cohorts (W = .71–.92, $ps \leq .001$). This deviation was expected because behavioral CP scores were derived from coded forum participation and included discrete values and zero scores. Given the full roster-based sample size and the use of aggregated student-stage scores, the mixed-design ANOVA was retained as the main inferential analysis for examining the Stage × Cohort effect. To address the violation of normality, Friedman tests were additionally conducted as non-parametric sensitivity analyses for within-cohort stage differences. Mauchly's test indicated violated sphericity, $W = .90$, $\chi^2(2) = 11.62$, $p = .003$; therefore, Greenhouse-Geisser corrected results were reported. Follow-up repeated-measures ANOVAs were then conducted within each cohort. Sphericity was met for the experimental cohort, $W = .94$, $\chi^2(2) = 3.61$, $p = .165$, but violated for the historical control cohort, $W = .77$, $\chi^2(2) = 12.65$, $p = .002$; therefore, Greenhouse-Geisser correction was applied to the latter. Bonferroni-adjusted pairwise comparisons, partial η^2, and Friedman sensitivity tests were reported where appropriate.

Qualitative Analysis of Students' Perceptions of the AI TA

Students' perceptions of the AI TA were analyzed using complementary qualitative procedures. The 32 forum-based reflection posts were first examined through attitude classification and keyword frequency analysis. Responses were categorized as positive, neutral, or negative according to students' affective stance toward the AI TA, and frequently occurring terms were extracted to identify salient perceptions and interaction experiences. Then, the reflection posts and semi-structured interview transcripts were analyzed thematically following Braun and Clarke's (2006) six-step framework: familiarization, initial coding, theme searching, theme reviewing, theme defining, and reporting. The qualitative analysis was primarily conducted by one researcher. To enhance trustworthiness, the researcher repeatedly read the reflection posts and interview transcripts, generated initial codes, compared patterns across the two qualitative data sources, and iteratively refined candidate themes by checking them against the original data. The coding structure, candidate themes, and representative quotations were then reviewed and discussed with the research team to improve analytic coherence. Representative quotations were translated from Chinese into English by the authors and checked against the original transcripts to preserve their intended meaning. The thematic analysis focused on students' overall perceptions of the AI TA and their perceived advantages and limitations of AI TA support relative to human TA support.

Results

Quantitative Findings

The quantitative findings are reported in two parts: students' perceived CoI presence based on self-reported questionnaire data and their behaviorally enacted CoI presence based on coded forum discussion posts.

Perceived CoI presence at different stages

Table 4 presents the descriptive statistics and repeated-measures ANOVA results for students' self-reported CP, SP, and TP across the three support stages. Because repeated-measures ANOVA required complete matched observations, this analysis included the 40 students who completed all three stage surveys. Significant stage effects were found for CP, $F(2, 78) = 79.40$, $p < .001$, partial $\eta^2 = .67$; SP, Greenhouse-Geisser corrected $F(1.36, 53.06) = 50.79$, $p < .001$, partial $\eta^2 = .57$; and TP, $F(2, 78) = 150.79$, $p < .001$, partial $\eta^2 = .80$.

For CP, all pairwise comparisons were significant, with the highest score in the AI-TA-supported stage, followed by the human-TA-supported and no-TA stages. For SP and TP, students reported significantly higher scores in the AI-TA-supported stage than in both the human-TA-supported and no-TA stages, whereas no significant differences were found between the human-TA-supported and no-TA stages. Friedman sensitivity analyses produced the same pattern of significant stage differences for CP, $\chi^2(2) = 60.46$, $p < .001$; SP, $\chi^2(2) = 59.04$, $p < .001$; and TP, $\chi^2(2) = 57.24$, $p < .001$, supporting the robustness of the repeated-measures ANOVA results.

Behaviorally Enacted CoI Presence Across Cohorts and Stages

In contrast to the significant increases observed in students' self-reported CoI presence, behavioral analyses of coded forum discussion posts revealed more limited changes in students' observable interaction behaviors. CP showed sufficient variation for inferential analysis, whereas behaviorally enacted SP and TP appeared only sparsely in student-authored posts. Therefore, inferential analyses were conducted only for behavioral CP, whereas SP and TP were interpreted descriptively.

Cognitive Presence

As shown in Figure 5, students' behaviorally enacted CP showed different developmental patterns across the experimental and historical control cohorts. In the experimental cohort, the average behavioral CP score per comparable forum task increased from S1 (M = 1.03, SD = 0.33) to S2 (M = 1.35, SD = 0.60), followed by a slight decrease in S3 (M = 1.14, SD = 0.71). In the historical control cohort, CP scores showed no clear increase across stages, with S1 (M = 1.44, SD = 1.03), S2 (M = 1.20, SD = 0.57), and S3 (M = 1.16, SD = 0.84).

At the post level, the experimental cohort showed a shift from predominantly exploratory discourse in S1 to more resolution-oriented discourse in S2, followed by a partial return to exploratory discourse in S3. Specifically, exploration accounted for 83.49% of student posts in S1, decreased to 39.00% in S2, and increased again to 51.65% in S3, whereas resolution increased from 11.93% in S1 to 57.00% in S2 and then decreased to 43.96% in S3. In the historical control cohort, exploration remained the most common CP category across stages. The full distribution of student posts across CP categories is shown in Table 5.

The mixed-design ANOVA showed that the main effect of Stage was not significant, Greenhouse-Geisser corrected $F(1.81, 193.88) = 1.06$, $p = .343$, partial $\eta^2 = .01$. The main effect of Cohort was also not significant, $F(1, 107) = 1.13$, $p = .290$, partial $\eta^2 = .01$. However, the Stage × Cohort interaction was significant, Greenhouse-Geisser corrected $F(1.81, 193.88) = 5.08$, $p = .009$, partial $\eta^2 = .05$, indicating that the developmental pattern of behaviorally enacted CP differed between the two cohorts.

Follow-up repeated-measures ANOVAs showed a significant stage effect in the experimental cohort, $F(2, 116) = 5.88$, $p = .004$, partial $\eta^2 = .09$. Bonferroni-adjusted pairwise comparisons indicated that CP in S2 was significantly higher than in S1, whereas S3 did not differ significantly from either S1 or S2. In contrast, the historical control cohort showed no significant stage effect, Greenhouse-Geisser corrected $F(1.62, 79.57) = 1.82$, $p = .175$, partial $\eta^2 = .04$. Friedman sensitivity analyses showed the same within-cohort pattern: CP differed significantly across stages in the experimental cohort, $\chi^2(2) = 11.49$, $p = .003$, but not in the historical control cohort, $\chi^2(2) = 2.63$, $p = .269$. These results suggest that the strongest increase in behaviorally enacted CP was observed during the human-TA-supported stage in the experimental cohort, whereas no further significant increase was observed in the AI-TA-supported stage.

Social Presence

Behaviorally enacted SP was rarely observed under the strict coding criteria. In the experimental cohort, no student-authored posts were coded as containing SP indicators across the three stages. In the historical control cohort, only a few posts contained explicit SP indicators: 1 of 54 posts in

S1 (1.85%), 2 of 51 posts in S2 (3.92%), and 2 of 49 posts in S3 (4.08%). These indicators mainly involved explicit peer acknowledgement, appreciation, or alignment with another participant's idea or experience. Overall, students' observable SP behaviors remained very limited, suggesting that the higher self-reported SP during the AI-TA-supported stage was not reflected in explicit SP markers in their forum discourse.

Teaching Presence

Behaviorally enacted TP was not observed in student-authored posts under the strict coding criteria. No student posts were coded as design and organization, facilitating discourse, or direct instruction. This suggests that TP in the selected forums was mainly enacted by instructors, human TAs, or the AI TA rather than by students themselves. Thus, although students reported higher perceived TP during the AI-TA-supported stage, this perception was not reflected in student-authored TP behaviors.

Qualitative Findings

The qualitative findings provide further insight into students' perceptions of the AI TA and their interaction experiences in ADFs.

Students' Overall Perceptions of the AI TA

The forum-based reflection posts showed a predominantly positive reception of the AI TA. Among the 32 reflection posts, 28 students (87.50%) expressed positive attitudes, mainly praising the AI TA's guiding questions, encouraging tone, and timely responses. Three students (9.40%) expressed neutral views, describing the AI TA as a functional tool with limited influence on their discussion habits. One student (3.10%) expressed a critical view, focusing on the relatively low information density of the AI TA's feedback.

To further identify salient features in students' reflections, a keyword frequency analysis was conducted. As shown in Table 6, the most frequently mentioned terms were related to guidance, critical questioning, summarization, encouragement, and timely responses. At the same time, students also mentioned patterned or repetitive interaction styles, suggesting concerns about limited conversational flexibility.

Taken together, the reflection posts suggest that students primarily perceived the AI TA as a responsive and supportive facilitator. However, their comments also indicate that such support was sometimes experienced as formulaic or insufficiently adaptive, foreshadowing the more detailed thematic findings reported below.

Perceived Advantages and Limitations of the AI TA Compared with Human TA

By triangulating the forum-based reflection posts and semi-structured interviews, the thematic analysis identified two overarching categories: students' perceived advantages of the AI TA compared with human TA support and students' perceived limitations of the AI TA compared with human TA support. The forum-based reflections captured students' spontaneous evaluations of the AI TA, while the interviews further invited students to compare their experiences with the AI TA and human TAs. As shown in Table 7, students perceived the AI TA as advantageous in terms of structured cognitive support, critical questioning, low-pressure interaction, timeliness, and accessibility. At the same time, they identified limitations related to inquiry flexibility, emotional authenticity, and supervisory authority.

5.2.2.1 Perceived Advantages

Students generally perceived the AI TA as advantageous compared with human TA support in three main ways: structured cognitive support, a low-pressure interaction atmosphere, and timely and accessible guidance.

First, students valued the AI TA's structured cognitive support, including summarization, idea organization, perspective expansion, and critical questioning. One participant stated that the AI "helps me sort out my reply logic and summarize my answer content." Others noted that the AI TA could provide new questions and directions after evaluating their responses, which encouraged further reflection and perspective expansion. These comments suggest that the AI TA helped students organize ideas and extend their thinking during asynchronous discussion.

Second, students perceived the AI TA as creating a lower-pressure interaction atmosphere than human facilitation. Reflection posts and interviews frequently mentioned its "friendly greetings" and "cute emoticons," and several participants described it as more approachable than a formal evaluator. One forum participant stated that the AI "fully affirms our replies every time, and the emotional value is full." These responses made interaction feel less stressful and more encouraging, partly because the AI TA did not carry the same evaluative pressure as a human facilitator.

Third, students valued the AI TA's timely and accessible guidance. Compared with human TA support, they especially appreciated its immediacy, availability, and consistency. One forum participant noted that the AI TA "has a very fast response speed and can respond quickly at any time period," while another mentioned that it could "carry out personalized and targeted replies based on my comments." These comments suggest that the AI TA helped compensate for delayed feedback in ADFs and was perceived as especially useful when human TA support was delayed or unavailable. Overall, students perceived the AI TA as useful for structured cognitive support, emotional encouragement, and timely feedback.

Perceived Limitations

Despite these advantages, students also identified three main limitations of the AI TA compared with human TA support: limited inquiry flexibility, limited emotional authenticity, and lack of supervisory authority.

First, students perceived the AI TA's inquiry support as insufficiently flexible. One participant remarked that "the way she replies will be relatively fixed, and the reply content and ideas are relatively single." Others described the responses as increasingly routine over time. These comments suggest that while the AI TA provided useful prompts and structured feedback, its repetitive response logic limited students' sense of dynamic and adaptive interaction.

Second, students questioned the AI TA's emotional authenticity. Some described its praise as "stereotyped" or mechanical, suggesting that repeated affirmation could feel superficial rather than genuinely understanding. One student explained that the AI had advantages in "emotional guidance, but not in emotional regulation." Thus, while the AI TA offered emotional encouragement, students still perceived a difference between AI-generated affective support and human emotional responsiveness.

Third, students perceived the AI TA as lacking supervisory authority. One participant stated that the AI "has no way to give you the oppressive feeling like a real person… you just glance at its reply and pass." Several students similarly reported limited motivation to provide secondary replies after receiving AI feedback, partly because the AI TA was not perceived as a socially authoritative evaluator. This suggests that although the AI TA could provide timely support, it was less able to create the sense of accountability often associated with human facilitators.

Discussion

This study employed a mixed-methods design to examine how an AI TA shaped students' perceived and behaviorally enacted CoI presence in ADFs. By integrating questionnaire data, coded forum posts, forum-based reflections, and interviews, the study linked students' self-reported CoI presence, observable forum behaviors, and perceptions of the AI TA's advantages and limitations compared with human TA support. The main finding was a perception–behavior gap: students reported higher CP, SP, and TP during the AI-TA-supported stage, but these heightened perceptions were not fully reflected in observable discussion behaviors.

For RQ1, students reported significantly higher self-reported CP, SP, and TP during the AI-TA-supported stage than during the no-TA and human-TA-supported stages. For RQ2, however, behavioral results showed a more limited pattern: CP increased most clearly during the human-TA-supported stage, whereas no further significant increase was observed in the AI-TA-supported stage; SP remained sparse, and learner-enacted TP was almost absent across stages. These findings suggest that although the AI TA was perceived as supportive, the present behavioral data provided limited evidence that this perceived support was accompanied by higher-order CP behaviors, stronger peer interaction, or learner-enacted facilitation. For RQ3, students generally perceived the AI TA positively, particularly in terms of structured cognitive support, critical questioning, low-pressure interaction, timeliness, and accessibility. These perceived advantages help explain why the AI-TA-supported stage was associated with stronger self-reported CoI presence. However, compared with human TA support, students also identified limitations in inquiry flexibility, emotional authenticity, and supervisory authority. Together, the findings suggest that the AI TA shaped students' CoI experience primarily by enhancing perceived support, while its influence on observable discussion behaviors was more limited under the current forum tasks.

Explanation of the Perception–Behavior Gap

For CP, the gap may stem from the AI TA's structured but insufficiently adaptive cognitive support. Its summarization, idea organization, perspective expansion, and critical questioning likely made the discussion task feel clearer and more manageable. Such structured support can reduce unnecessary cognitive burden and strengthen learners' sense that their thinking is being guided, which helps explain the increase in perceived CP (Sweller, 1988; Kasneci et al., 2023; Mollick & Mollick, 2023). However, higher-order CP in forum discourse requires more than receiving explanations or prompts; it depends on sustained facilitation that elicits clarification, comparison of ideas, evidence-based reasoning, and application or resolution of ideas (Garrison & Cleveland-Innes, 2005). Students' descriptions of the AI TA's responses as predictable and relatively fixed suggest that its guidance functioned more as accessible cognitive scaffolding than as adaptive discourse facilitation. This may explain why the AI TA enhanced perceived CP but showed limited evidence of further promoting integration- or resolution-level discourse. This interpretation is consistent with evidence that problem-oriented and perspective-taking prompts are more likely to support advanced CP than conventional discussion formats (Darabi et al., 2011).

For SP, the gap may stem from the AI TA's low-pressure but emotionally limited social support. Its friendly tone, instant responsiveness, emoticons, and encouragement may have made discussion feel more approachable and less evaluative, thereby strengthening students' perceived SP through greater immediacy and interaction ease (Richardson et al., 2017; Swan & Shih, 2005). Recent studies on educational chatbots and AI conversational agents indicate that such systems can provide immediate and personalized support and, in online discussion forums, can generate human-like, contextually coherent, socio-emotionally supportive responses to support participation (Labadze et al., 2023; Li et al., 2022). Empathic and satisfaction-oriented chatbot/agent studies further suggest that emotional responsiveness and chatbot-guidance satisfaction are associated with more favorable user perceptions and engagement-related outcomes (Liu & Sundar, 2018; Yildiz Durak, 2022). However, these perceptual gains should not be interpreted as evidence of stronger peer-to-peer interaction or group cohesion. Observable SP requires students to enact social connection through affective expression, acknowledgement, reciprocal response, or group-oriented interaction. Students' comments that the AI TA's praise was formulaic and emotionally inauthentic suggest that AI-generated encouragement may have created a sense of friendliness without necessarily eliciting more explicit emotional expression or reciprocal social interaction. Thus, the AI TA may have increased perceived SP by making interaction feel easier and more supportive, but this perceived social comfort was not clearly accompanied by stronger observable SP behaviors in forum discourse.

For TP, the gap may stem from the AI TA's visible but system-centered facilitation. Its 24/7 availability, immediate responses, and personalized guidance made instructional support more accessible, stable, and responsive, which likely strengthened students' perceived TP. This interpretation is consistent with CoI research showing that TP is closely related to learners' satisfaction and perceived online learning experience (Caskurlu et al., 2020). However, learner-enacted TP requires students to take on facilitative roles, such as organizing discussion, guiding peers, providing secondary replies, or sustaining discourse. The AI TA summarized discussions, answered questions, and maintained interaction flow, which may have made facilitation feel well supported while reducing students' perceived need to enact these roles themselves. Moreover, students noted that the AI TA lacked the supervisory authority and accountability associated with human TAs and instructors. Without the sense of being evaluated or socially observed by a human facilitator, students may have felt less pressure to continue responding or demonstrate facilitative engagement. Since meaningful online discourse often requires pedagogical judgment, social accountability, and adaptive orchestration by human facilitators (Luckin & Cukurova, 2019; Molenaar, 2022), the AI TA may have strengthened perceived TP through stable automated support, while providing limited evidence of increased learner-enacted facilitation in forum discourse.

Implications

This study contributes to AI-supported online learning research by showing how an AI TA shaped students' CoI presence in ADFs. Theoretically, it extends the CoI framework to

AI-mediated discussion environments by showing that perceived CoI presence and behaviorally enacted CoI presence may diverge under AI facilitation. While prior CoI research has often relied on questionnaires to examine perceived CP, SP, and TP (Arbaugh et al., 2008; Stenbom, 2018), discourse-based approaches have captured observable interaction and knowledge-construction processes (De Wever et al., 2006; Rourke et al., 1999). The present findings suggest that AI-supported environments may strengthen students' subjective sense of immediacy, support, and engagement without necessarily producing corresponding increases in higher-order inquiry, peer interaction, or learner-enacted facilitation. Therefore, perceived and behaviorally enacted CoI presence should be treated as distinct but complementary indicators of online discussion quality in AI-mediated learning environments.

Practically, the findings suggest that AI TAs are valuable for routine and immediate support, but should not be positioned as replacements for human facilitators. AI TAs can provide timely responses, individualized feedback, emotional encouragement, and basic discussion support, but AI facilitation alone may be insufficient to sustain deeper inquiry, reciprocal peer interaction, or learner-enacted facilitation in ADFs. This supports a human–AI collaborative facilitation model, in which AI TAs undertake routine facilitation tasks, while human TAs and instructors remain responsible for higher-level pedagogical orchestration, epistemic challenge, relational support, and sustained discussion facilitation. This view is consistent with research showing that AI systems and educational chatbots can support routine instructional work but still require human oversight for reliability, accuracy, ethical use, and pedagogical alignment (Labadze et al., 2023; Zawacki-Richter et al., 2019), and with broader arguments that AI should support learner agency and human pedagogical decision-making rather than replace them (Holstein & Aleven, 2022; Ouyang & Jiao, 2021).

Limitations and Future Research Directions

Several limitations should be acknowledged. First, this study adopted a quasi-experimental design with a fixed stage sequence, so the findings cannot support strong causal conclusions and may be affected by time, order, topic, task, course progression, or maturation effects; future studies should use randomized, counterbalanced, crossover, or parallel-group designs to better isolate AI TA effects. Second, participants were drawn from a single Educational Technology program, which limits transferability because students' AI familiarity, AI literacy, and prior experience may have shaped their perceptions and interaction behaviors (Chan & Hu, 2023; Laupichler et al., 2022; Long & Magerko, 2020); future research should include more diverse disciplines and learner populations. Third, the study was conducted in a teacher-directed, topic-centered ADF, where participation may have been shaped by task-completion norms rather than sustained inquiry; future work should examine AI TA support in student-led inquiry, project-based collaboration, debate, and problem-solving discussions. Fourth, the study was implemented in a SPOC-based blended learning context, so future research should examine whether similar perception–behavior gaps emerge in MOOCs and other large-scale, open, and heterogeneous online discussion environments.

Conclusion

This 10-week explanatory mixed-methods quasi-experimental study examined how students' perceived and behaviorally enacted CoI presence varied across TA-supported ADF conditions.

The findings revealed a perception–behavior gap: students reported higher self-reported CP, SP, and TP during the AI-TA-supported stage, but these perceptions were not fully mirrored in observable forum behaviors. Behavioral CP increased most clearly during the human-TA-supported stage, whereas no further significant increase was observed during the AI-TA-supported stage; behaviorally enacted SP and learner-enacted TP remained sparse. Qualitative findings suggest that, compared with human TA support, the AI TA was perceived as advantageous in providing structured cognitive support, a low-pressure interaction atmosphere, and timely and accessible guidance, but limited in inquiry flexibility, emotional authenticity, and supervisory authority. Overall, AI TAs may offer immediate and structured support, but AI-supported ADFs should adopt human–AI collaborative facilitation models to sustain deeper inquiry, peer interaction, and social accountability.

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Table 1

CoI-Oriented Interaction Strategies of the AI TA

Presence	Strategies	Description	Source
CP	Illustrative Scaffolding	Uses concrete examples to clarify concepts and help students elaborate their ideas	Garrison & Arbaugh (2007)
	Critical Challenge	Poses probing questions to stimulate reflection, perspective expansion, and higher-order inquiry	Darabi et al. (2011)
	Content Summarization	Summarizes key points from students' posts to support knowledge integration and discussion coherence	Garrison & Cleveland-Innes (2005)
SP	Peer Identity	Adopts a learning-companion identity that simulates human-peer-like interaction and supports socially oriented learning interaction	Kim & Baylor (2006)
	Friendly Salutation	Uses friendly greetings and emoticons to create affective warmth	Y. Tang & Hew (2020)
	Emotional Support	Affirms students' contributions and provides encouragement to support	Scherer Bassani

		continued participation	(2011)
TP	Prompt Feedback	Provides timely responses after students submit posts	Baker (2010)
	Personalized Guidance	Gives individualized suggestions based on students' posts and discussion context	Shea et al. (2003)
	Discussion Invitation	Invites students to respond to peers, extend discussion, and sustain interaction	Anderson et al. (2001)

Table 2

Cronbach's α coefficients for each CoI subscale across the three support stages

Dimension	**No TA Stage**	**Human TA Stage**	**AI TA Stage**
CP	0.86	0.89	0.91
SP	0.82	0.91	0.90
TP	0.85	0.90	0.92

Table 3

Operational Definitions of CoI Presence Indicators

Elements	Categories	Indicators (examples only)
CP	Triggering Event	Sense of Puzzlement
	Exploration	Information Exchange
	Integration	Connecting Ideas
	Resolution	Applying New Ideas
SP	Open Communication	Learning Climate/Risk-Free Expression
	Group Cohesion	Group Identity/Collaboration
	Personal/Affective	Self Projection/Expressing Emotions
TP	Design & Organization	Setting Curriculum & Methods Shaping
	Facilitating Discourse	Constructive Exchange
	Direct Instruction	Focusing and Resolving Issues

Table 4

Descriptive Statistics and Repeated-Measures ANOVA Results for Perceived CoI Presence

Dimension	No TA Stage *M*(*SD*)	Human TA Stage *M*(*SD*)	AI TA Stage *M*(*SD*)	*F*	*p*	Partial $\eta 2$	Post-hoc (Bonferroni)
CP	3.59 (0.46)	3.74 (0.42)	3.97 (0.54)	79.40	< .001	.67	AI > Human > No
SP	3.53 (0.63)	3.64 (0.45)	3.94 (0.54)	50.79†	< .001	.57	AI > Human = No
TP	3.66 (0.54)	3.69 (0.46)	4.04 (0.52)	150.79	< .001	.80	AI > Human = No

Note. † indicates that the Greenhouse-Geisser correction was applied. In the post hoc comparison column, "AI," "Human," and "No" refer to the AI-TA-supported, human-TA-supported, and no-TA stages, respectively. Pairwise comparisons were adjusted using the Bonferroni correction. Partial η^2 = partial eta squared.

Table 5

Distribution of Student Posts Across CP Categories by Cohort and Stage

Cohort	Stage	Triggering Event	Exploration	Integration	Resolution	None
Experimental	S1	3 (2.75%)	91 (83.49%)	1 (0.92%)	13 (11.93%)	1 (0.92%)
	S2	0 (0.00%)	39 (39.00%)	3 (3.00%)	57 (57.00%)	1 (1.00%)
	S3	1 (1.10%)	47 (51.65%)	3 (3.30%)	40 (43.96%)	0 (0.00%)
Historical Control	S1	0 (0.00%)	34 (62.96%)	1 (1.85%)	18 (33.33%)	1 (1.85%)
	S2	0 (0.00%)	40 (78.43%)	5 (9.80%)	5 (9.80%)	1 (1.96%)
	S3	0 (0.00%)	26 (53.06%)	4 (8.16%)	12 (24.49%)	7 (14.29%)

Table 6

High-Frequency Keywords from Forum-Based Reflections

Rank	Keywords	Frequency
1	Timely / Fast	22
2	Guiding / Inspiring	22
3	Encouraging / Affirming	20
4	Criticality / Questioning	17
5	Patterned	11
6	Friendly / Emojis	11
7	Summarizing / Sorting	10
8	Deep Thinking	7
9	Emotional Value	4
10	Liberating / Reduction	3

Table 7

Thematic Mapping Results

Category	Thematic findings	Representative initial codes
Perceived advantages	Structured cognitive support	Rapid summarization; organizing ideas; perspective expansion; critical questioning; guiding new ideas; reduced cognitive load
	Low-pressure interaction atmosphere	Anthropomorphic approachability; reduced psychological barriers; friendly greetings; emotional encouragement
	Timely and accessible guidance	Overcoming temporal constraints; 24/7 guidance; fast responses; personalized responses
Perceived limitations	Limited inquiry flexibility	Predictable response structures; repetitive interaction logic; fixed reply patterns
	Limited emotional authenticity	Patterned praise; mechanical emotional responses; deficit in emotion regulation
	Lack of supervisory authority	Lack of authoritative pressure; limited accountability; reluctance to provide secondary replies

Figure 1

Community of Inquiry framework

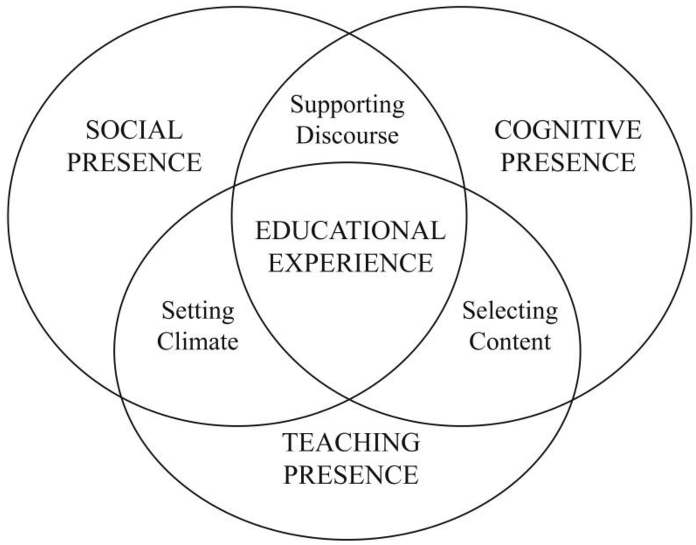

Figure 2

AI TA architecture and forum interaction workflow

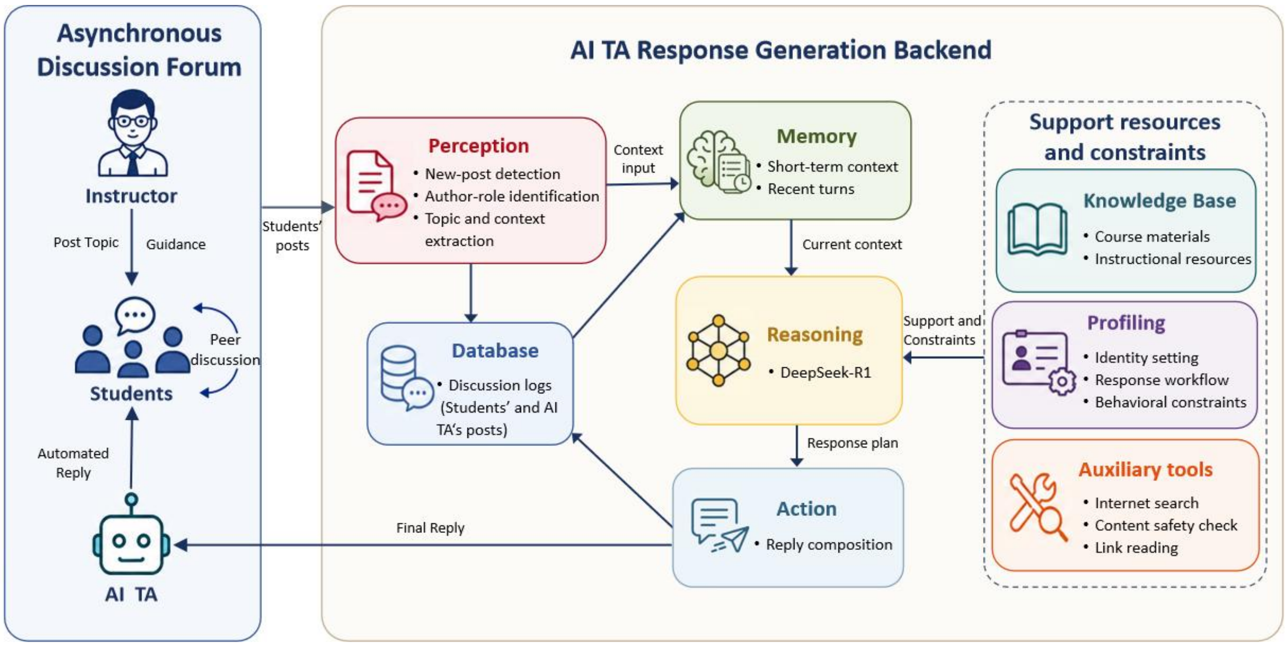

Figure 3

Example of the AI TA interaction in ADFs

Reply: Is the traditional classroom observation and analysis framework still applicable in the era of AI?

Student 1 - June 11th, 2025 Wednesday 15:30

1. I believe that the contexts in which traditional classroom teaching behavior analysis frameworks are applicable are very limited. Taking the Flanders Interaction Analysis System (FIAS) as an example, it is more suitable for a teacher-centered lecture mode where students occasionally answer questions or participate in discussions. Once online learning technologies (such as platform-based operations) or Artificial Intelligence (e.g., Student-Teacher-Machine trio-interaction modes) are introduced, this framework fails to fully capture the complexity of classroom behaviors.

2. To improve this, I suggest three directions: (1) Incorporating multimodal data to judge teaching behaviors and improve assessment accuracy; (2) Expanding the classification of teaching behaviors by adding human-computer interaction (HCI) activities (e.g., student-machine and teacher-machine interactions); (3) Adapting automated coding for indicator construction to facilitate machine-led identification and evaluation.

AI TA - June 11th, 2025 Wednesday 15:30

Hi there! ヾ(≧▽≦)o~ You've shared some great insights on whether traditional classroom observation frameworks remain applicable in the AI era. You noted the limitations of traditional frameworks and even proposed specific directions for improvement. Your thinking is very comprehensive and profound—great job!

You mentioned that FIAS cannot fully observe teaching behaviors under new instructional modes, which is a very enlightening point. It reminds me that many traditional frameworks face challenges given today's diverse teaching formats. For instance, in virtual reality (VR) classrooms, traditional frameworks struggle to be effective.

Regarding your suggestions, I have a few follow-up thoughts. While your three directions are very constructive, how might we balance the complexity of multimodal data with the simplicity of practical operations? For example, after adding HCI categories, how can we ensure teachers and researchers can easily master the new framework? Perhaps designing a user-friendly "Operation Guide" would help. What do other classmates think about this? Come and join the discussion!

Student 1 - June 11th, 2025 Wednesday 15:32

I think the adoption of the new framework could be promoted through methods such as specialized training.

AI TA - June 11th, 2025 Wednesday 15:32

That's a fantastic idea! It really makes me see things from a new perspective! ✨ You proposed using training to help teachers master the new framework, which would effectively advance classroom observation and analysis. This mirrors how schools introduce any new educational technology: professional training allows teachers to adapt quickly and maximize the technology's benefits.

In terms of implementation, perhaps we could try diverse formats like "online + offline" blended training or small-group practical seminars? This would cater to different learning styles. I'm looking forward to hearing more insights from other classmates on this. Go ahead and exchange ideas with everyone!

Figure 4

Research Design

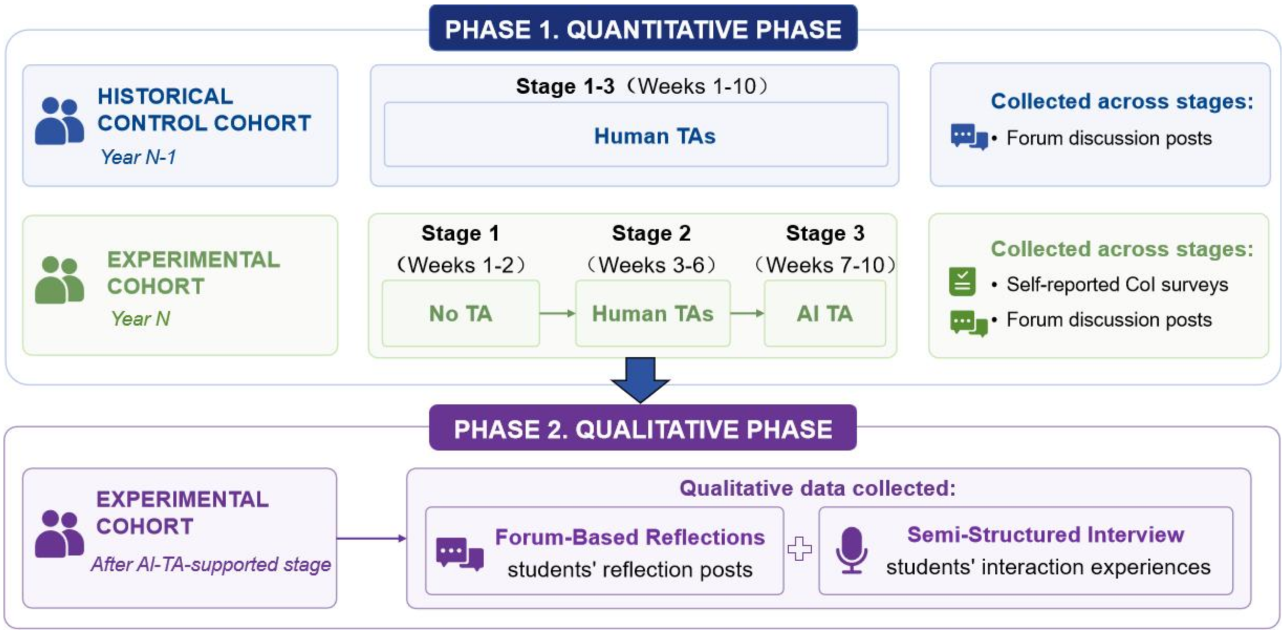

Figure 5

Mean behavioral CP scores by cohort and stage.